Lightweight image segmentation for echocardiography

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Abstract

Accurate segmentation of the left ventricle in echocardiography can enable fully automatic extraction of clinical measurements such as volumes and ejection fraction. nnU-Net is a self-adapting framework that dynamically configures training pipelines for U-Nets based on the dataset, and is considered state-of-the-art in medical image segmentation. While models configured by nnU-Net perform well, they are large and slow, thus limiting real-time utility. Our objective is to investigate the key components for achieving robust segmentation in echocardiography, and to develop a significantly smaller and faster model that maintains the accuracy of nnU-Net.

We identified the most impactful components of nnU-Net through an ablation study. Starting with a baseline U-Net, we evaluated nnU-Net's major design choices by incrementally introducing individual components and assessed their impact on segmentation performance. The investigated components include data augmentation schemes, architectural modifications, loss function variants, and post-processing techniques.

The selected components were deployed on a lightweight U-Net about 16 times smaller than the default nnU-Net configuration. This model achieved segmentation performance that was not significantly different ($p > 0.05$) from nnU-Net on the CAMUS dataset (N=500 patients). Generalization ability was evaluated on an independent internal dataset (N=311).

1 Introduction

Cardiovascular disease (CVD) is the leading global cause of death, responsible for 20.5 million lives lost in 2021 [2, 9]. Analyzing 2D echocardiographic images allows for extraction of clinical measurements like ejection fraction and left ventricular volume, which assess cardiovascular health. Manual or semi-automatic segmentation is today daily work [7]. Fully automatic cardiac segmentation could reduce manual workload and free up clinician time for patient care.

nnU-Net, introduced in 2018, automatically configures U-Net pipelines without manual tuning and achieves state-of-the-art performance across various datasets [5]. New architectures such as Transformers [12] and Mamba [3] have been adapted to the medical imaging domain, and many claim to outperform CNN-based U-Nets [1, 4, 10]. The creators of nnU-Net show that CNN-based U-Nets still give the best performance for most medical imaging datasets if tuned well, and advocate that future research should prioritize thorough validation over introducing novel architectures [6].

This work conducts an ablation study to identify the key performance-driving components of nnU-Net. To this end, nnU-Net is first picked apart and a series of experiments are run where various components are disabled. Then, the components are added to and evaluated on a lightweight U-Net. A smaller model is desirable because it trains faster, offers better generalization to new datasets and gives faster inference time which is particularly useful when deployed on portable ultrasound scanners with limited compute for real-time inference.

2 Method

To bridge the performance gap between nnU-Net and a lightweight U-Net on the CAMUS dataset, we incrementally add components from nnU-Net to a smaller U-Net. Successful additions showing a performance increase are retained for downstream experiments. Models are evaluated using a per-class Dice coefficient and Hausdorff distance, and a count of anatomical outliers. Statistical significance is tested with the Wilcoxon signed-rank test. For modifications beyond configuration files, custom `trainer` classes are implemented to modify components such as loss functions, sampling methods, and augmentation schemes.

2.1 Datasets

2.1.1 CAMUS

The CAMUS dataset [7] consists of 2D echocardiographic sequences from 500 patients, with annotations for the left ventricle (LV), myocardium (MYO), and left atrium (LA). It includes 2000 image-annotation pairs, each 256×256 pixels, captured using GE Vivid E95 scanners.

2.1.2 HUNT4

The HUNT4Echo dataset [11] originates from a large-scale health study in Trøndelag, including 2462 LV-focused echocardiographic exams. Among these, 311 exams provide segmentation labels for LV, MYO, and LA

2.1.3 Differences

Although both HUNT4 and CAMUS present the same clinical views, the two datasets cannot be directly combined and jointly trained on due to differences in annotation conventions and imaging techniques. In CAMUS, the myocardium is annotated significantly thicker than in HUNT4, illustrated in Figure 1. Additionally, HUNT4-images have less variance and are regarded as more clinically accurate [13]. CAMUS is used for most training experiments in this work due to it being publically available.

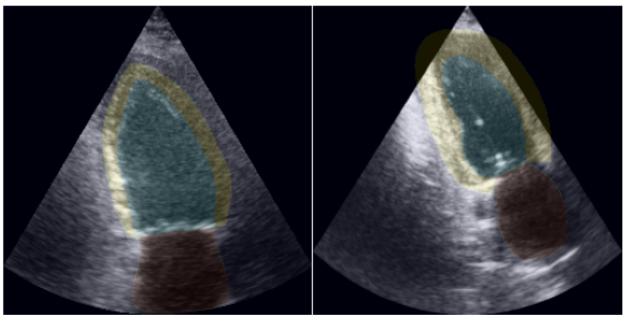


Figure 1: Ground truth segmentations from HUNT4 (left) and CAMUS (right). Images from HUNT4 are consistently LV-focused, while those from CAMUS are not.

2.1.4 Dataset Usage and Splits

CAMUS is used for development, testing, and tuning, while HUNT4 is used to evaluate the models ability to generalize across different imaging conditions. An 80/10/10 train/validation/test split is applied to both datasets, yielding 1600 training, 200 validation, and 200 test images for CAMUS, and 846 training, 105 validation, and 105 test images for HUNT4.

2.2 Baselines

Two baselines are established. One for a lightweight vanilla U-Net similar to U-Net 1 [7] and another for nnU-Net [5], see Table 1 for a comparison. The goal of this work is to analyze the nnU-Net by incrementally adding its components to the U-Net 1 baseline in order to find the most important parts in terms of performance. All experiments using the lightweight U-Net follow the this training scheme:

- 100 epochs with early stopping (patience=50)
- Adam optimizer with initial learning rate of 0.001
- Batch size 32

For a fair comparison, the nnU-Net model is also trained for 100 epochs. This is unproblematic because empirical observations show convergence well before this point.

2.3 Experiments

Several components of nnU-Net are systematically analyzed through incremental experiments:

1. Normalization scheme: Instance normalization (IN) and Batch normalization (BN).
2. Activation functions: ReLU, Leaky ReLU, Mish, and GELU.
3. Connected component analysis (CCA) and the morphological operations opening and closing are used to exploit the fact that the images in CAMUS and HUNT4 only contain one instance of each target structure.
4. Data augmentation following the scheme outlined in Table 2, implemented using Albumentations[1]
5. Loss functions: Dice and average of Dice and Cross Entropy.
6. Deep Supervision (DS) was proposed in [8] and used by [15, 16] to train U-Nets. 1×1 convolutions reduce the number of channels down to the number of classes. Then, the feature maps are upscaled using nearest neighbor interpolation, and finally softmaxed. A new loss function is chosen to accommodate Deep supervision: $L = L_{\text{final}} + \sum_{i=1}^{n} \lambda_i L_{\text{auxiliary}_i}$, where λ is set to 0.3, L_{final} is the loss between the final prediction and the ground truth, and $L_{\text{auxiliary}_i}$ is the loss between the upscaled intermediate feature map i and the ground truth.
7. Three additional networks with varying sizes, shown in Table 3, are trained. The goal is to find the needed capacity of a model to learn the distribution of the training set.
8. Architectural changes: Residual connections and transposed convolutions.

3 Results

3.1 Baselines

The two baseline models are trained following the configuration in Table 1, with metrics as shown in Table 4. The U-Net 1 baseline predicts $\frac{42}{200}$ anatomical outliers, while the nnU-Net baseline achieves an average dice score of 0.89 and average Hausdorff distance of 8.54 after 100 epochs.

[1] Albumentations Documentation

Architecture	Number of channels	Lowest resolution	Upsampling scheme	Downsampling scheme	Normalization scheme	Batch size	Activation function	Loss function	# Trainable parameters	Augmentations
U-Net 1	$32 \downarrow 128 \uparrow 16$	8×8	Nearest neighbor interpolation	2×2 Maxpool	-	32	ReLU	Dice	2M	-
nnU-Net	$32 \downarrow 512 \uparrow 32$	4×4	Transposed convolutions	Strided convolutions	InstanceNorm	49	Leaky-ReLU	Dice+CE	33M	Rotating and scaling, Gaussian noise, Gaussian blur, Brightness, Contrast, Low-res sim, Gamma and Mirroring

Table 1: Architectural differences between the two baselines. Number of channels indicate the number of channels at the first, deepest and last feature map.

Config	Shift	Scale	Rot.	Gamma	Noise	Blackout	Bri. Con. Sat.	Hue
I	$(-0.05, 05)$	$(-0.1, 0.05)$	$(-5, 5)$	$(90, 110)$	$(5, 10)$	0.25	$(0.95, 1.05)$	$(-0.05, 0.05)$
II	$(-0.10, 10)$	$(-0.2, 0.1)$	$(-10, 10)$	$(85, 115)$	$(10, 25)$	0.25	$(0.90, 1.10)$	$(-0.10, 0.10)$
III	$(-0.15, 15)$	$(-0.3, 0.15)$	$(-15, 15)$	$(80, 120)$	$(20, 35)$	0.25	$(0.85, 1.15)$	$(-0.15, 0.15)$
IV	$(-0.20, 20)$	$(-0.4, 0.20)$	$(-20, 20)$	$(70, 130)$	$(30, 45)$	0.25	$(0.80, 1.20)$	$(-0.20, 0.20)$
V	$(-0.25, 25)$	$(-0.5, 0.25)$	$(-30, 30)$	$(60, 140)$	$(40, 50)$	0.25	$(0.75, 1.25)$	$(-0.25, 0.25)$
Affine	$(-0.1, 0.1)$	$(-0.2, 0.1)$	$(-10, 10)$	-	-	-	-	-

Table 2: Data augmentation experiment plan. Blackout is a rectangle with sides $\sim \mathcal{U}(25, 65)$ px placed randomly. Other parameters are inputs to Albumentations-augmentations.

Arch.	Channels (start↓end↑final)	Low Res.	BS	Scheduler	# Params
U-Net 1	$32 \downarrow 128 \uparrow 16$	8×8	32	-	2M
U-Net 2	$64 \downarrow 512 \uparrow 32$	8×8	16	ReduceLROnPlateau	11M
U-Net 3	$64 \downarrow 1024 \uparrow 64$	16×16	8	ReduceLROnPlateau	31M
U-Net 0	$16 \downarrow 64 \uparrow 16$	64×64	64	ReduceLROnPlateau	119K

Table 3: Architectural configurations used to evaluate the effect of network size. Normalization scheme, loss function, and augmentations are consistent across models.

3.2 Experiments on nnU-Net

Disabling the entire data augmentation scheme results in the Dice scores 0.92, 0.85, 0.88 and Hausdorff distances 10.01, 10.36, 10.92. The difference in both metrics compared to the baseline are statistically significant with $p < 0.05$, using the Wilcoxon signed-rank test, except the Dice score for the LA class. With deep supervision enabled, the difference in metrics compared to the baseline is not statistically significant compared to only disabling data augmentations, indicating that for nnU-Net, augmentations are more important that deep supervision.

Next, the connected component analysis done by nnU-Nets postprocessing engine is investigated. For the baseline, postprocessing does not improve performance significantly. For the model trained without augmentations, both metrics improve for LV and MYO, but not for LA. The model trained without augmentations or deep supervision is statistically significantly worse for all classes on both metrics with $p < 0.05$ when postprocessing is enabled. The largest discrepancy is in the Hausdorff distance, which is likely because it measures the maximum distance, and is thus more prone to outliers, which the postprocessing helps reduce.

Metric	U-Net 1	nnU-Net
Dice LV	0.90	0.93
Dice MYO	0.82	0.86
Dice LA	0.86	0.89
Hausdorff LV	10.13	7.10
Hausdorff MYO	12.87	9.21
Hausdorff LA	12.04	9.31
# Anatomical Outliers	42	6

Table 4: Baseline metrics.

3.3 Building a lightweight, performant U-Net

Starting from the baseline U-Net 1, Table 6 summarizes the results for the top-performing variation for each addition outlined in 2.3. Components showing performance improvements are carried forward for subsequent runs. Next, we show detailed results from experiments on select components.

Data augmentation

Table 5 shows the results from running experiments with the configurations in Table 2. They show that, out of the tested configurations, a moderate augmentation scheme - one applying only affine transformations, is optimal. It results in a significantly higher Dice score and lower Hausdorff distance across all classes ($p < 0.05$). In contrast, the nnU-net-like augmentations (Config I-V) lead to the lowest number of outliers, but achieve worse Dice scores and Hausdorff distances.

Config	Dice LV	Dice MYO	Dice LA	Haus. LV	Haus. MYO	Haus. LA	# Anatomical Outliers
Baseline	0.90	0.82	0.86	10.13	12.87	12.04	42
Baseline w/ BN	0.92	0.84	0.87	8.40	11.21	10.47	26
I	0.92	**0.85**	**0.88**	8.38	11.13	9.93	12
II	0.92	**0.85**	**0.88**	8.22	10.96	10.56	**10**
III	0.92	0.84	0.87	8.57	11.13	10.72	**10**
IV	0.91	0.83	0.87	9.46	12.67	10.94	14
V	0.82	0.58	0.69	27.27	55.12	26.17	66
Affine	**0.93**	**0.85**	**0.88**	**7.80**	**10.00**	**9.91**	12

Table 5: Dice score and Hausdorff distance for U-Net 1 with different augmentation configurations

Deep supervision

Introducing deep supervision to the baseline U-Net with batch normalization yields a statistically significant improvement in the Hausdorff distance for the

LV, while performance for other metrics and labels is worse. When affine augmentations are combined with DS, both evaluated metrics are significantly improved relative to training without deep supervision.

Upscaled feature maps from models trained with and without DS are visualized in Figure 2. The intermediate decoder blocks of the model trained with DS produce more anatomically consistent segmentation maps compared to those generated without.

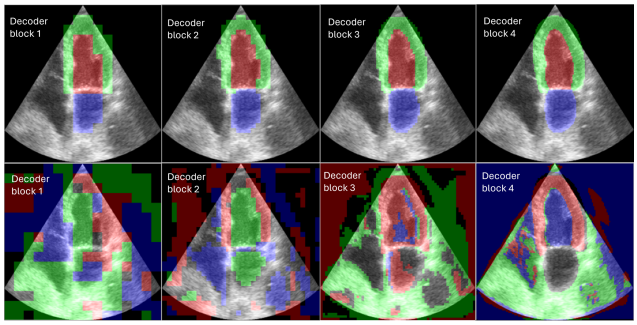


Figure 2: Upscaled intermediate feature maps: top row shows results with deep supervision, bottom row without. Maps from early decoder layers appear pixelated due to an upscaling factor of 16.

Addition	Dice LV	Dice MYO	Dice LA	Haus. LV	Haus. MYO	Haus. LA	Conclusion
Baseline	0.90	0.82	0.86	10.13	12.87	12.04	-
Normalization	0.91	0.84	0.86	10.08	11.64	12.38	use BN
Activation functions	0.92	0.84	0.87	8.97	12.27	11.15	use MISH
Postprocessing	0.92	0.84	0.87	8.40	11.21	10.47	use CCA
Data augmentation	0.93	0.85	0.88	7.80	10.00	9.91	use affine
Loss function	0.93	0.85	0.89	7.75	10.46	9.58	use avg(Dice, CE)
Deep supervision	0.93	0.85	0.89	7.49	9.93	9.22	apply
Size	0.93	0.85	0.88	8.22	10.52	9.54	discard
Architecture	0.92	0.85	0.88	8.10	10.38	9.62	discard

Table 6: Summary of metrics for each anatomical structure for experiments on U-Net 1. Conclusion indicates which additions are kept for downstream experiments.

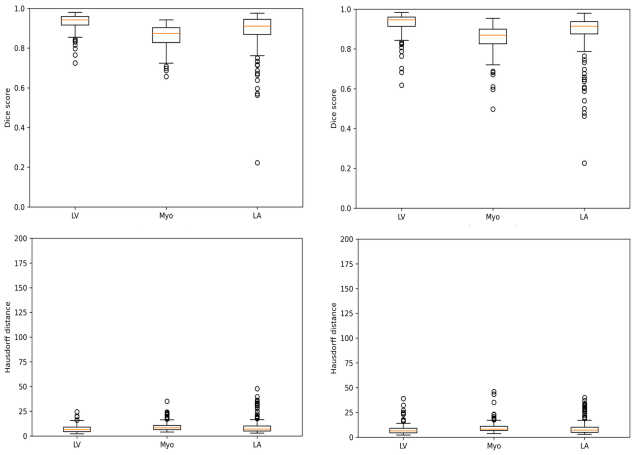
3.4 The final model

The final lightweight model extends U-Net 1 by integrating all components showing an increase in segmentation performance. The model is summarized in Table 7. It achieves the Dice scores $0.93, 0.85, 0.89$ and the Hausdorff distances $7.49, 9.93, 9.22$ for the LV, MYO and LA respectively.

Figure 3 presents boxplots of both metrics for the lightweight U-Net and the nnU-Net baseline. The Wilcoxon signed-rank test [14] shows no statistically significant difference between the two models, with $p = 0.87$ for Dice score and $p = 0.56$ for Hausdorff distance. This indicates that there is no difference in median performance.

3.5 Results on HUNT4

TODO: If there is space.



(a) nnU-Net (b) Lightweight U-Net

Figure 3: Boxplots of Dice scores (top) and Hausdorff distances (bottom) for nnU-Net and our lightweight U-Net.

Component	Description
Model	U-Net 1
Downsampling scheme	2×2 Maxpool
Upsampling scheme	Nearest neighbor interpolation
Normalization scheme	Batch normalization
Postprocessing	Connected component analysis
Data augmentation	Shift(0.1), Scale(-0.2, 0.1), Rotate(10)
Loss function	Dice + Cross Entropy (avg.)
Activation function	Mish
Deep supervision	Yes, $\lambda = 0.3$ for all aux. losses

Table 7: Details of the final model.

4 Discussion

In this work we use select components from the nnU-Net pipeline to create a lightweight U-Net with similar performance to nnU-Net-trained models on the CAMUS dataset.

The most important components are normalization layers and data augmentation, with the latter being the most impactful. Notably, although augmented images may not appear clinically realistic (Figure **??**), they still enhance performance—highlighting that augmentation design should prioritize metric improvements over visual plausibility.

Experiments show that the full nnU-Net augmentation scheme, while effective for large models, is excessive for a lightweight U-Net. Simpler affine transformations—shifting, scaling, and rotation—yield the best results, significantly reducing anatomical outliers (from 26 to 12), and improving Dice scores and Hausdorff distances. In contrast, augmentations like Gamma adjustment, Gaussian noise, and ColorJitter introduce unnecessary noise and degrade performance.

Combining affine augmentations with deep supervision further boosts performance with minimal added complexity. Deep supervision helps guide the decoder's intermediate layers toward anatomically plausible shapes (Figure **??**), improving interpretability and accelerating convergence. This aligns with prior findings [**?**], confirming its effectiveness for ultrasound seg-

mentation.

In summary, for smaller models, a simplified augmentation scheme—focused on affine transforms—paired with deep supervision, offers the best balance of performance and model clarity.

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